Netaji Subhash Engineering College

PRESENTATION ON :-

Slide 1: A cover page containing the student's name, roll number, etc.
Slides 2 to 4: Your content
Slide 5: One case study/use case

DISTRIBUTED SYSTEMS PEC-IT601B(6590)

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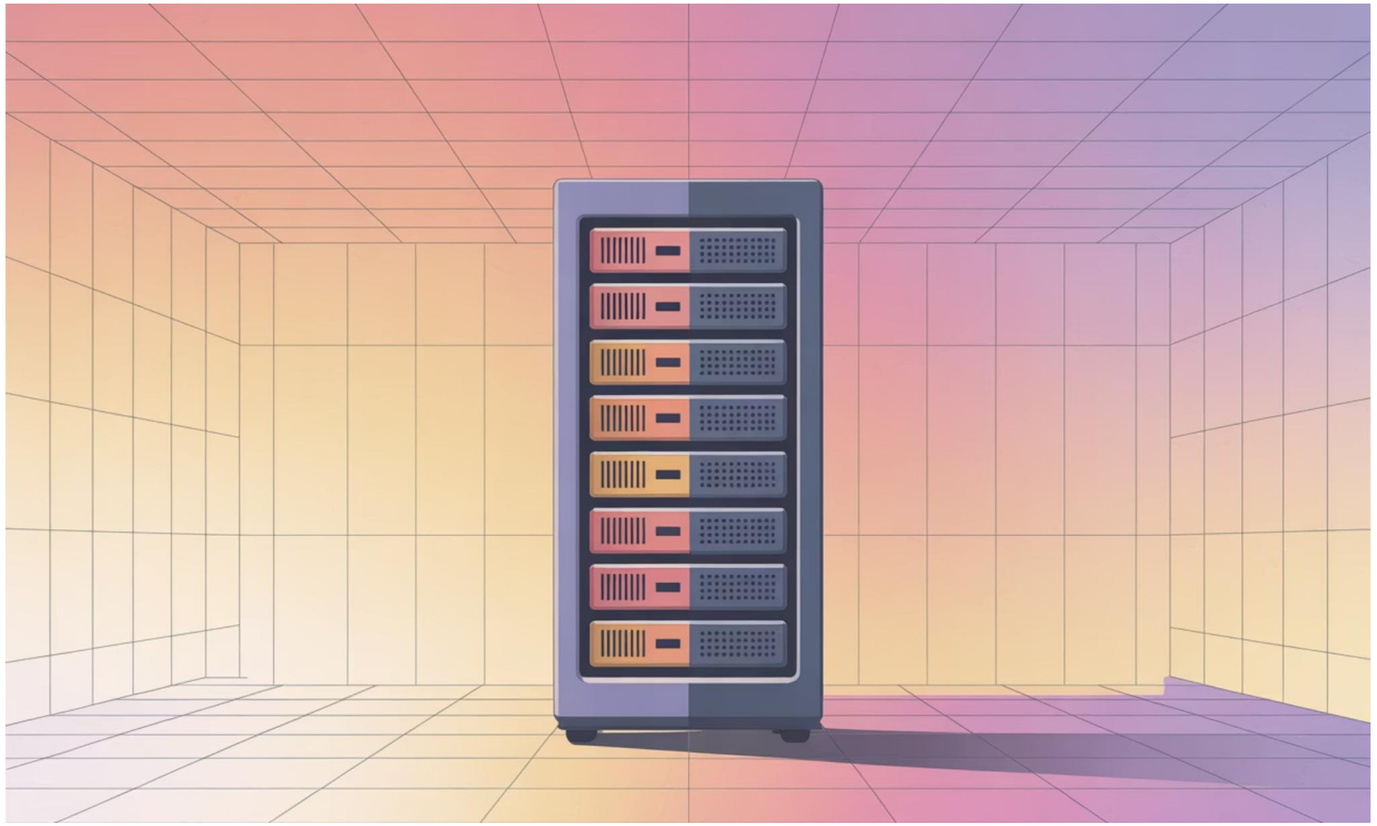
What is a Centralized DBMS?



A **Centralized Database Management System** stores and manages all data at a single physical location, typically on one server or within a single data center. This consolidated approach simplifies various aspects of data handling.

Advantages

- Easier to design and maintain, reducing initial complexity.
- Strong data consistency and ACID (Atomicity, Consistency, Isolation, Durability) compliance are inherent.
- Lower initial cost and simpler security enforcement due to a single point of control.

Disadvantages

- Single point of failure risks significant downtime and data loss.
- Scalability is limited by the server's capacity, leading to vertical scaling constraints.
- Performance bottlenecks can occur under heavy user load, impacting responsiveness.

What is a Distributed DBMS?

A **Distributed Database Management System** (DDBMS) stores data across multiple interconnected nodes, which can be located in different physical or geographic locations. This architecture supports horizontal scaling and enhanced data availability.

Advantages

- High availability and fault tolerance, as there's no single point of failure.
- Improved performance through parallel processing of queries and distributed workloads.
- Geographic data distribution reduces latency for global users, enhancing user experience.

Disadvantages

- Complex design and maintenance, requiring sophisticated management strategies.
- Significant challenges in maintaining strong data consistency across all distributed nodes.
- Higher initial setup and ongoing operational costs due to increased infrastructure and complexity.

Key Differences Between Centralized and Distributed DBMS

Feature	Centralized DBMS	Distributed DBMS
Data Location	Single physical location	Multiple interconnected locations
Scalability	Limited (vertical scaling via hardware upgrade)	High (horizontal scaling by adding nodes)
Fault Tolerance	Low (single point of failure leads to outage)	High (redundancy and replication ensure uptime)
Performance	Can degrade significantly under heavy load	Improved speed through parallel processing and local access
Complexity	Simpler to manage and administer	More complex (consistency, synchronization, and recovery)
Cost	Lower initial and operational costs	Higher initial setup and ongoing operational costs

Case Study: Global E-Commerce Platform Using Distributed DBMS

Leading e-commerce platforms like **Amazon** rely heavily on distributed database architectures to manage their vast and dynamic data. This approach is critical for serving millions of users worldwide efficiently.

Challenge

Serve a global user base with ultra-low latency and continuous high availability, regardless of geographic location.

Solution

Implementation of a highly distributed database architecture with multiple data centers strategically placed across continents.

Benefits

- Data replication ensures zero downtime even during regional server failures.
- Geographic data partitioning minimizes access latency, speeding up transactions.
- System easily scales horizontally to handle massive transaction volumes during peak sales events.

Result

A robust, fast, and scalable platform capable of supporting Amazon's extensive global operations and customer base.

Conclusion

1

Centralized DBMS

Ideal for small to medium-sized applications where simplicity, strong consistency, and lower costs are priorities.

2

Distributed DBMS

Excels in large-scale, high-availability, and geographically dispersed environments demanding scalability and low latency.

The choice between a centralized and distributed DBMS critically depends on the application's specific scale, performance requirements, fault tolerance needs, and budget constraints. Understanding these core differences empowers informed architectural decisions for effective modern data management strategies.